

INPUT: Physical System
PDEs, ODEs, Device Physics Equations

STAGE 1: Ψ -HDL PIPELINE

Section III.A

1. Teacher Network Training (Eq. 2)

- Minimizes physics residuals
- Maximizes data fidelity
- Multi-objective loss function

2. Student Distillation & Clustering (Eq. 14)

- Knowledge distillation from teacher
- Hardware complexity penalty for compression
- Hierarchical agglomerative clustering (HAC)

OUTPUT 1: Cluster Centers (μ_k^ℓ) & Relation Matrices (R^ℓ)
Up to ~99.6% parameter-count compression (best case)

STAGE 2: HARDWARE SPECIFICATION EXTRACTION

Section III.B

Computational Complexity
TOPS_{req} via throughput analysis
Safety margin $\gamma = 1.2\text{--}1.5\times$ throughput

Memory Estimation
 $M_{\text{total}} = \text{Weights} + \text{Activations} + \text{State}$
Conservative: $\beta = 2\times$ memory overhead

Energy / Power
 $P_{\text{avg}} = P_{\text{dyn}} + P_{\text{static}}$
 $E_{\text{inf}} = E_{\text{compute}} + E_{\text{memory}} + P_{\text{static}} \cdot T_{\text{inf}}$

Platform Recommendation Algorithm

TinyML (very low throughput)

STM32H7, Nordic nRF52840

Mid-Range (few TOPS-class NPUs)

TI AM62A, TI TDA4VM

High-Performance (tens of TOPS+)

Jetson Orin Nano, Hailo-8

STAGE 3: SOLAR-CONSTRAINED TRAINING & VALIDATION

Section III.C

Solar Power Availability Model

$$P_{\text{solar}}(t) = P_{\text{max}} \cdot \max\left(0, \sin\left(\frac{\pi(t-6)}{12}\right)\right) \cdot W(t)$$

GPU Power (assumed) 250 W • PV intensity 48 gCO₂/kWh • Example target d_{cycle} ≈ 50% (with appropriate sizing/calibration)

Regime 1: Continuous (Grid)

- Baseline reference
- Standard ω

Regime 2: Passive ($\kappa = 0$)

- Checkpoint only (optimizer state)
- Preserves Adam momentum across interruptions

Regime 3: Active ($\kappa > 0$)

- Adaptive regularization (κ -mechanism)
- **Marginal gain, Burgers only: -8.5% to -11.2%** (passive is default)

► Budget-Sensitivity (C→B, Table IV)

Spans three orders of magnitude across seven PDEs (5 of 7 resolved at 95% CI)

Burgers -3.2% • Laplace +41.5% • Allen-Cahn +97.6% • Heat +176.8% • Advection +620.0%

No clean predictor from PDE class or temporal coupling

OUTPUT: Sustainable Deployment Strategy

Compressed Model:

Physics-consistent, cluster-based

Optimal Hardware:

Right-sized platform selection

Verilog-A HDL:

Circuit-ready compact model

96.8% Cost Savings:

\$8 MCU vs. \$249 Jetson module

~18× Carbon Reduction:

224.7 kg/device (99.9% hardware-led)

No Accuracy Penalty:

Burgers PDE; varies by budget sensitivity